

COURIER MANAGEMENT SYSYEM (CMS)

User Story:

Courier and logistics organizations can face difficulty when shipment and delivery activities depend on manual work and disconnected records. Without a centralized tracking mechanism, parcel movement may not be visible in a timely manner. Manual agent assignment can create scheduling conflicts, while limited notification and reporting facilities can reduce operational visibility.

The Courier Management System (CMS) addresses these concerns by bringing customer, shipment, delivery agent, branch, invoice and reporting activities into one Salesforce-based platform.

1. INTRODUCTION

1.1 Project Overview:

The Courier Management System is designed to digitize major courier activities. It provides a central place for shipment booking, tracking, delivery-agent assignment, status updates, invoicing and operational reports. Salesforce Lightning, objects, flows, validation rules and dashboards are used to organize these functions.

1.2 Purpose:

The purpose of CMS is to reduce fragmented courier operations and improve visibility across branches. The system supports faster shipment processing, consistent data entry, controlled access and useful management reports.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template):

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain
PS-3	Delivery Agent	receive clear assignments	allocation is manual	information changes	confused

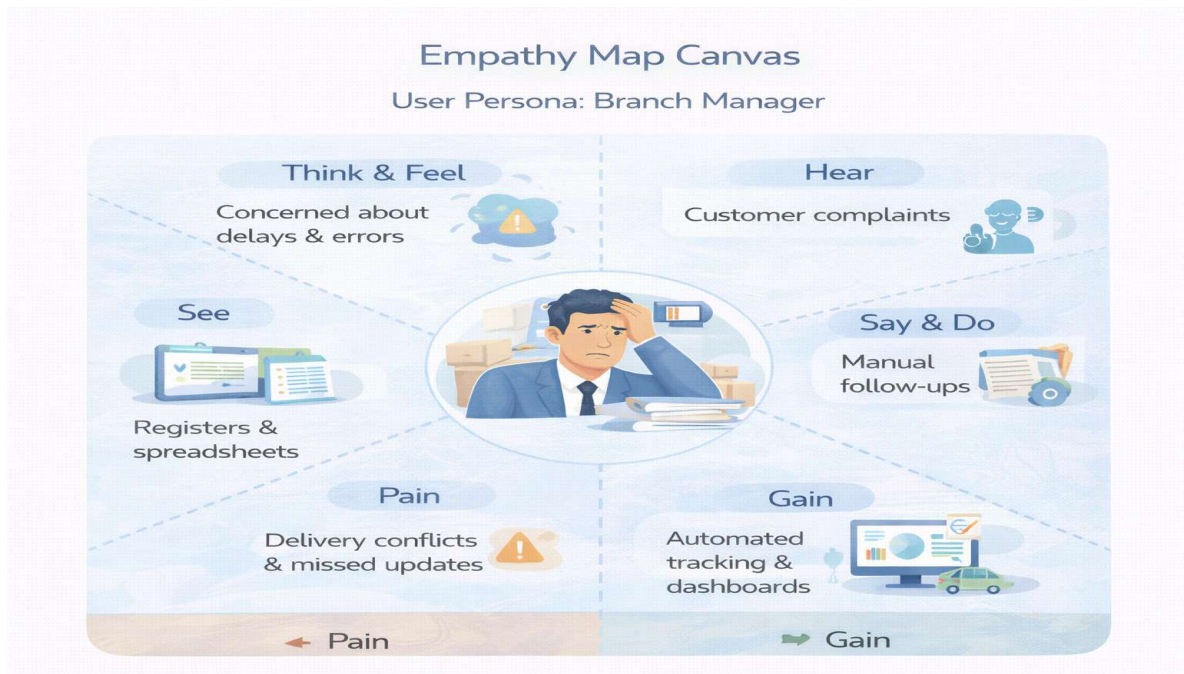
Problem Statements



2.2 Empathy Map Canvas:

User Persona: Branch Manager

- **Think & Feel:** Concerned about delays, errors and branch workload.
- **Hear:** Customer questions and operational complaints.
- **See:** Registers, spreadsheets and multiple records.
- **Say & Do:** Manual follow-ups.
- **Pain:** Delivery conflicts and missed updates.
- **Gain:** Automated tracking and dashboards.



2.3 Brainstorming & Idea Prioritization:

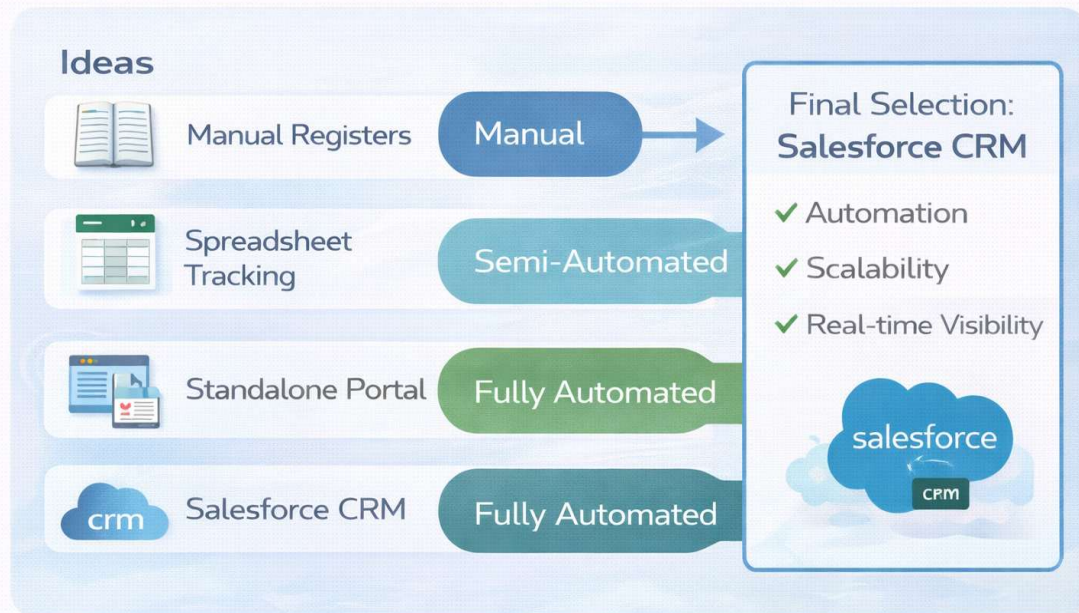
Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection

- Automation
- Scalability
- Centralized data
- Real-time visibility

2.3 Brainstorming & Idea Prioritization – Courier Management System



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map:

Stages: Booking → Shipment Creation → Agent Assignment → Transit → Delivery
Update → Invoice → Reporting

3.1 Customer Journey Map – Courier Management System



3.2 Solution Requirements:

Functional Requirements:

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports & dashboards

Non-Functional Requirements:

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Consistent automation
Performance	Fast tracking updates
Availability	Cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram:

Customer Input → Shipment Creation → Validation → Agent Assignment →
Delivery Update → Invoice → Reports

3.3 Data Flow Diagram – Courier Management System



3.4 Technology Stack:

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows

Security	Profiles, Roles
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit:

CMS brings fragmented courier activities into a centralized Salesforce platform. Shipment tracking, delivery assignment, billing and reporting are organized around related objects.



4.2 Proposed Solution

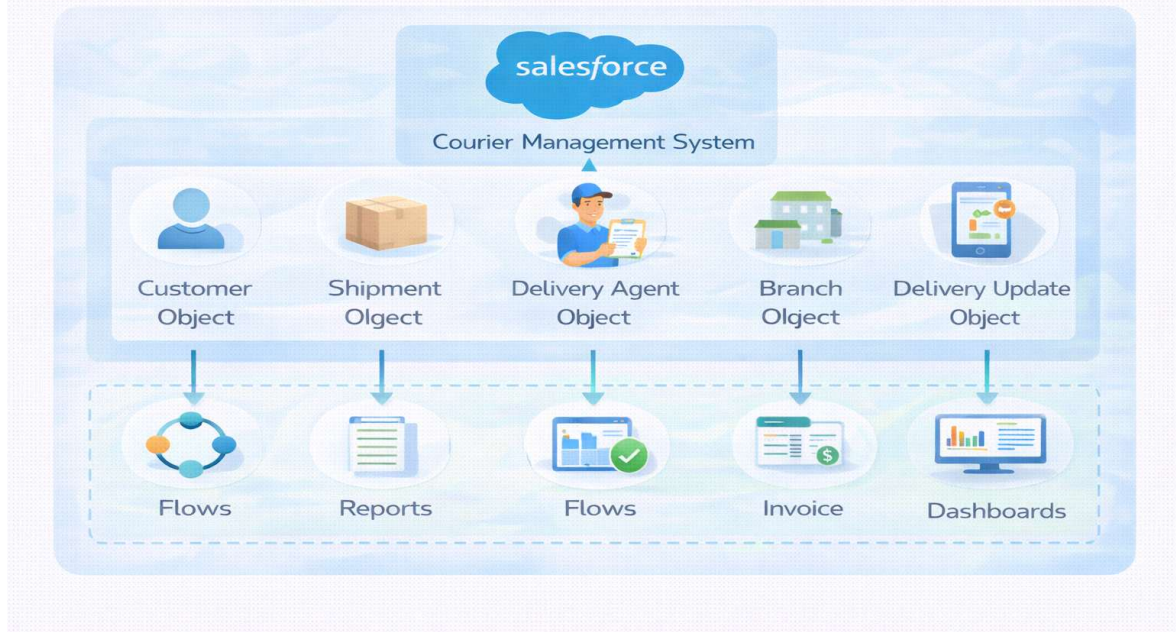
Function requirements:

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking and billing
Impact	Faster and organized processing
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

4.3 Solution Architecture – Courier Management System



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning:

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation for estimation

Product Backlog, Sprint Schedule, and Estimation:

Sprint	Requirement	User Story / Task	Points	Priority
Sprint-1	Developer Setup	Configure Salesforce developer account	3	High
Sprint-2	Data Modeling	Create CMS objects and relationships	5	High
Sprint-2	Data Modeling	Create tabs and Lightning App	5	High
Sprint-3	Automation	Validate shipment information	3	High
Sprint-3	Automation	Automate shipment status	3	High
Sprint-4	Security	Configure role-based access	5	High
Sprint-5	Reports	Create operational reports	4	High
Sprint-6	Dashboards	Create management dashboard	4	Medium

Project Tracker, Velocity & Burndown Chart:

Sprint	Story Points	Duration	Planned Start	Planned End
Sprint-1	20	6 Days	Week 1	Week 1
Sprint-2	20	6 Days	Week 2	Week 2
Sprint-3	20	6 Days	Week 3	Week 3
Sprint-4	20	6 Days	Week 4	Week 4

6. Project Development Phase

The development phase covers Salesforce account setup, custom objects and relationships, page layouts, validation rules, automation flows, application configuration, security, reports, dashboards and testing.

What you'll learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships
- Creating page layouts & record types
- Writing validation rules
- Building automation with flows
- Developing reports & dashboards
- Setting up security with profiles, permission sets and sharing
- End-to-end application testing

Milestone 1: Salesforce Account

Introduction

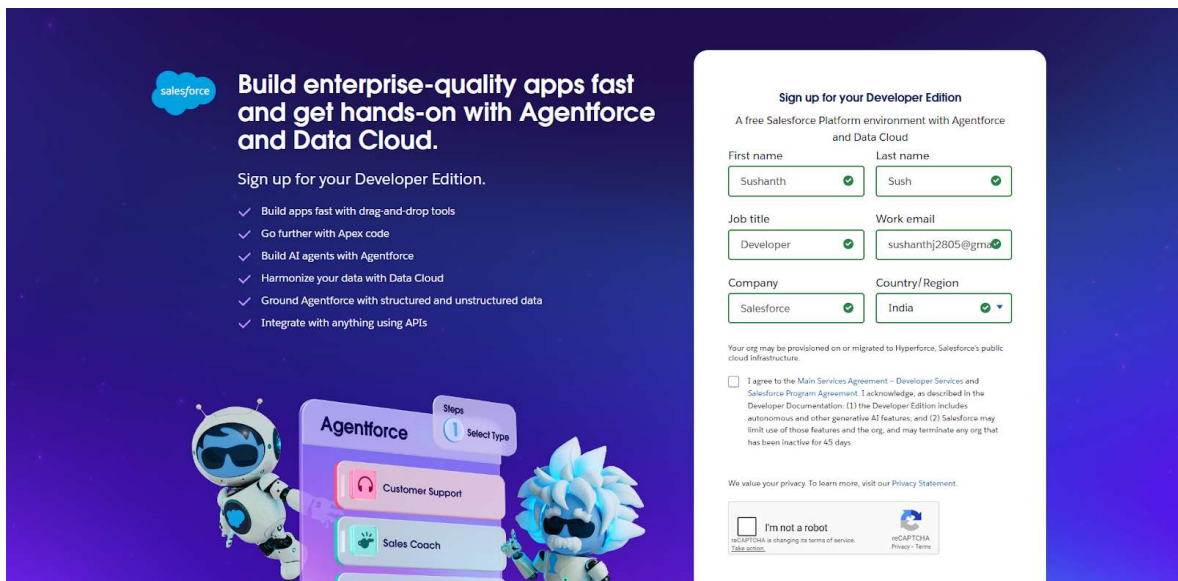
Salesforce provides cloud-based tools for organizing customer and operational data. In this project, it is used as the foundation for courier workflows.

What Is Salesforce?

Salesforce is a cloud customer platform providing tools for customer relationships, business data, automation, analysis and collaboration.

Activity 1: Creating Developer Account

- Open the Salesforce Developer signup page.
- Enter first name, last name and email.
- Enter job title as Developer.
- Enter college or organization name.
- Select India and submit the registration form.



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First name: Sushanth ✓ Last name: Sush ✓

Job title: Developer ✓ Work email: sushanthj2805@gmail.com ✓

Company: Salesforce ✓ Country/Region: India ✓

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

☐ I agree to the Main Services Agreement - Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

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☐ I'm not a robot
reCAPTCHA is changing its terms of service.
2024-05-06

reCAPTCHA
Privacy - Terms

Activity 2: Account Activation

- Open the registration email.
- Verify the account.

Milestone 2- Object

What is an object?

Salesforce objects work like structured database tables. They contain fields and records for storing application data.

Custom Objects Created

- Customer__c – customer personal and contact information
- Shipment__c – parcel, source, destination and tracking information
- Delivery_Agent__c – delivery agent details
- Branch__c – courier branch offices
- Delivery_Update__c – shipment status updates
- Invoice__c – billing information

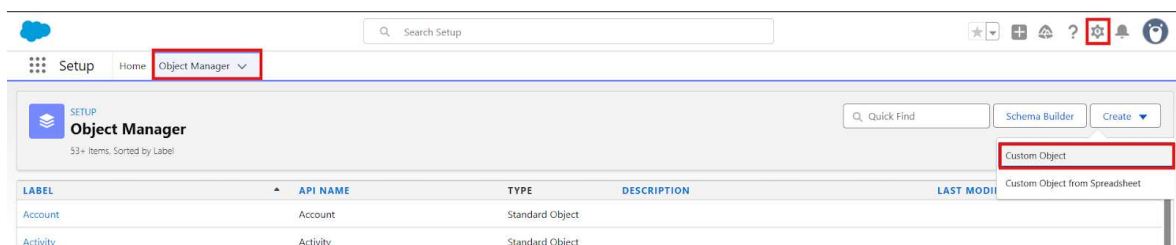
Relationships

- Customer → Shipment (Lookup)
- Shipment → Delivery_Agent (Lookup)
- Delivery_Agent → Branch (Lookup)
- Delivery_Update → Shipment (Master-Detail)
- Invoice → Shipment (Lookup)

Steps to Implement

Activity 1: Create Customer Object

- Setup → Object Manager → Create → Custom Object



- Label: Customer
- Plural Label: Customers
- Record Name: Customer Name (Text)
- Enable reporting and field history where required
- Save

The screenshot shows the 'Optional Features' section of the Salesforce Custom Object Settings for 'Customer'. The 'Allow Reports', 'Allow Activities', and 'Track Field History' checkboxes are checked and highlighted with a red box. Below this, the 'Object Classification' section shows that the object is classified as an Enterprise Application object. The 'Deployment Status' section shows 'Deployed' is selected. The 'Search Status' section shows 'Allow Search' is checked and highlighted with a red box. The 'Object Creation Options' section shows 'Add Notes and Attachments related list to default page layout' is checked. At the bottom, the 'Save' button is highlighted with a red box.

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

- ☒ Add Notes and Attachments related list to default page layout
- ☐ Launch New Custom Tab Wizard after saving this custom object

[Save](#) [Save & New](#) [Cancel](#)

Activity 2: Create Shipment Object

- Label: Shipment
- Plural Label: Shipments
- Record Name: Tracking Number (Auto Number)
- Example Format: SHP-{000000}
- Enable reports and field history
- Save

Activity 3: Create Delivery_Agent Object

- Label: Delivery Agent
- Plural Label: Delivery Agents
- Record Name: Agent Name (Text)

- Enable reports and field history
- Save

Activity 4: Create Branch Object

- Label: Branch
- Plural Label: Branches
- Record Name: Branch Name (Text)
- Enable reports
- Save

Activity 5: Create Delivery_Update Object

- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)
- Example Format: UPD-{0000}
- Enable reports and field history
- Save

Activity 6: Create Invoice Object

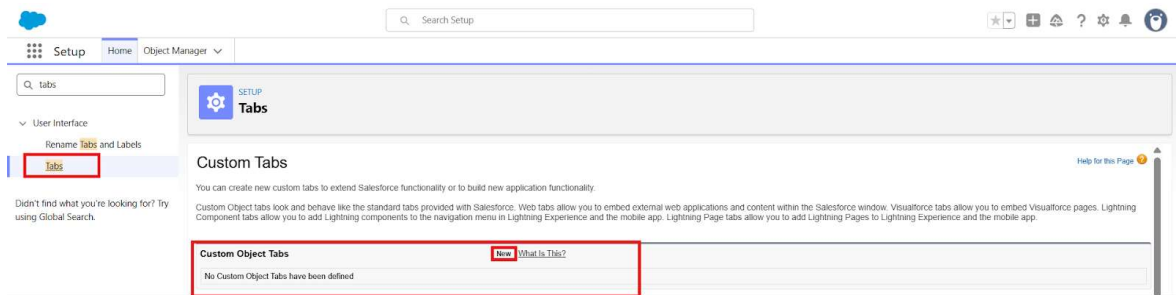
- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)
- Example Format: INV-{0000}
- Enable reports and field history
- Save

Milestone 3 - Tabs

Tabs in Salesforce help users access object data and application information quickly.

Activity: Creating a Tab for Customer Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
-



- Select the required object.
- Choose a tab style.
- Keep suitable default settings.
- Click Next and Save.

Activity: Creating a Tab for Shipment Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
- Select the required object.
- Choose a tab style.
- Keep suitable default settings.
- Click Next and Save.

Activity: Creating a Tab for Delivery Agent Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
- Select the required object.
- Choose a tab style.
- Keep suitable default settings.

- Click Next and Save.

Activity: Creating a Tab for Branch Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
- Select the required object.
- Choose a tab style.
- Keep suitable default settings.
- Click Next and Save.

Activity: Creating a Tab for Delivery Update Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
- Select the required object.
- Choose a tab style.
- Keep suitable default settings.
- Click Next and Save.

Activity: Creating a Tab for Invoice Object

- Open Setup and search for Tabs.
- Under Custom Object Tabs, select New.
- Select the required object.
- Choose a tab style.
- Keep suitable default settings.
- Click Next and Save.

Milestone 4: Fields & Relationships

What are fields?

Fields represent individual data columns within an object and store values required for records.

Activity 1: Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes
Customer Type	Customer_Type__c	Picklist	Yes
City	City__c	Text	No
State	State__c	Text	No

Pincode	Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

Activity 2: Creation of fields for the Shipment Object

Field	API Name	Data Type	Required
Tracking Number	Name	Auto Number	Yes
Customer	Customer__c	Lookup(Customer)	Yes
Source Address	Source_Address__c	Text Area	Yes
Destination Address	Destination_Address__c	Text Area	Yes
Weight	Weight__c	Number	Yes
Price	Price__c	Currency	Yes
Status	Status__c	Picklist	Yes
Expected Date	Expected_Date__c	Date	Yes

Assigned Agent	Assigned_Agent__c	Lookup(Delivery Agent)	No
Shipment Type	Shipment_Type__c	Picklist	Yes

Status values: Booked, In Transit, Out for Delivery, Delivered, Cancelled. Shipment Type values: Domestic, International.

Activity 3: Creation of fields for Delivery_Agent Object

Field Name	API Name	Data Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

Status values: Active, Inactive, On Leave.

Activity 4: Creation of fields for Branch Object

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

Activity 5: Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes

Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

Status values: In Transit, Out for Delivery, Delivered, Delayed.

Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status values: Paid, Unpaid, Pending.

Relationships

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Milestone 5-Validation Rules

Validation rules check information before a record is saved and prevent invalid or incomplete data.

Validation Rule	Formula	Error Message
Email Must Not Be Blank	ISBLANK(Email__c)	Email is required.
Phone Must Be 10 Digits	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Pincode Must Be 6 Digits	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Weight Greater Than Zero	Weight__c <= 0	Weight must be greater than zero.
Price Greater Than Zero	Price__c <= 0	Price must be greater than zero.

Expected Date Not in Past	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Valid Agent Email	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.
Branch Code Required	ISBLANK(Branch_Code__c)	Branch Code is mandatory.
Remarks Required for Delay	AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))	Remarks are required when status is Delayed.

Implementation Steps

Open Setup → Object Manager → select the object → Validation Rules → New. Enter rule name, formula, error message and location, check syntax, activate the rule and save.

Milestone 6 - Page Layouts

Page Layouts control fields, sections, buttons and related lists displayed on a record page.

Edit Page Layout for Customer__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Customer Information: Customer Name, Customer Type, Email, Phone, Is Active. Address Details: Address, City, State, Pincode.

Edit Page Layout for Shipment__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Shipment Details: Tracking Number, Customer, Shipment Type, Weight, Price, Status. Address Information: Source Address, Destination Address. Delivery Information: Expected Date, Assigned Agent.

Edit Page Layout for Delivery_Agent__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Agent Details: Agent Name, Email, Phone, Branch, Status, Joining Date. Additional Information: Related Shipments, Related Delivery Updates.

Edit Page Layout for Branch__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Branch Information: Branch Name, Branch Code, City, State, Manager Name.

Edit Page Layout for Delivery_Update__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Update Information: Update Number, Shipment, Status, Update Date, Remarks.

Edit Page Layout for Invoice__c

Open Setup → Object Manager → Page Layouts → open the default layout. Add sections, arrange the fields and save. Invoice Details: Invoice Number, Shipment, Amount, Invoice Date, Payment Status.

Milestone 7 - Flows

A Salesforce Flow automates business processes through a visual interface.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

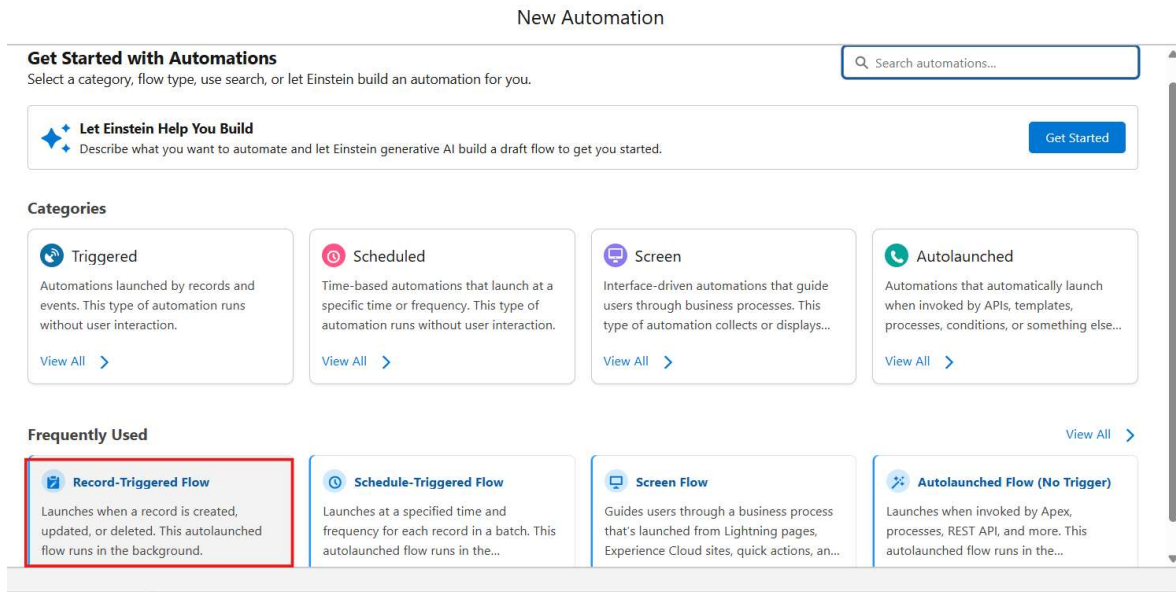
Whenever a Delivery_Update__c record is created, the related Shipment__c status is updated with the latest delivery status.

Part 1: Create the Record-Triggered Flow

- Open Setup → Flows → New Flow.
- Select Record-Triggered Flow.
- Object: Delivery_Update__c.



- Trigger: A record is created.
- Condition: None (always run).
- Optimize for Actions and Related Records.



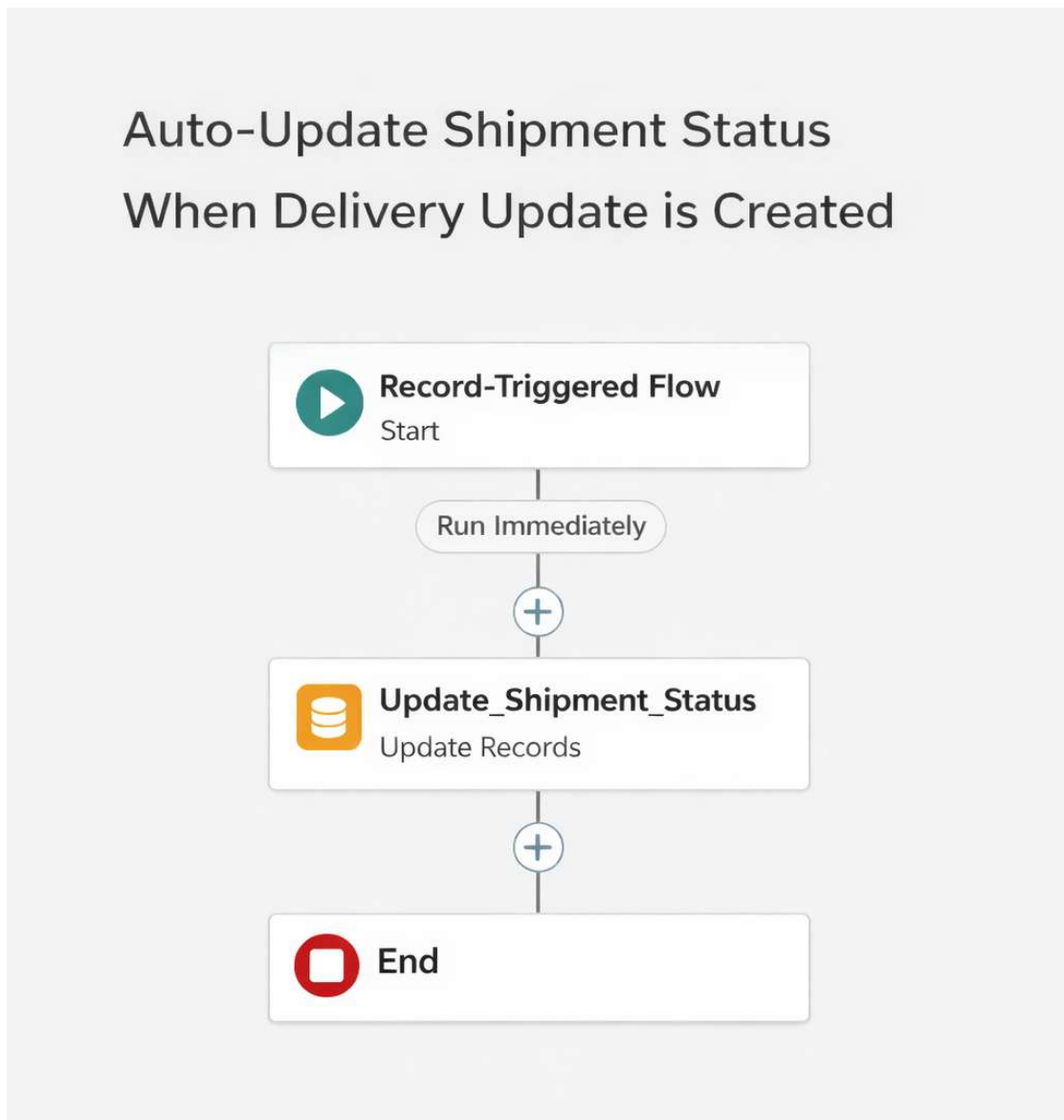
Part 2: Update Related Shipment Record

- Add Update Records.
- Label: Update_Shipment_Status.
- Find Shipment__c by Id = {!\$Record.Shipment__c}.
- Set Shipment Status__c = {!\$Record.Status__c}.

Part 3: Save and Activate

- Save as Auto_Update_Shipment_Status.
- Activate the flow.
- Confirm Active status.

Final flow layout:



Milestone 8 - The Lightning App

What is an App?

A Salesforce app groups related tabs and functions into one working environment.

Activity 1: Create a Lightning App

- Setup → App Manager → New Lightning App.

- App Name: Courier Management System.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ

* Developer Name ⓘ

Description ⓘ

App Branding

Image ⓘ



Clear

Primary Color Hex Value ⓘ

▼

Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

Next

- Add an icon if required.
- Add Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice, Reports and Dashboards.
- Select appropriate profiles.
- Save & Finish.

New Lightning App

User Profiles

Choose the user profiles that can access this app.

Available Profiles

- Analytics Cloud Integration User
- Analytics Cloud Security User
- Anypoint Integration
- Authenticated Website
- Authenticated Website
- B2B Reordering Portal Buyer Profile

Selected Profiles

System Administrator

Save & Finish

Milestone 9- Profile

What is a profile?

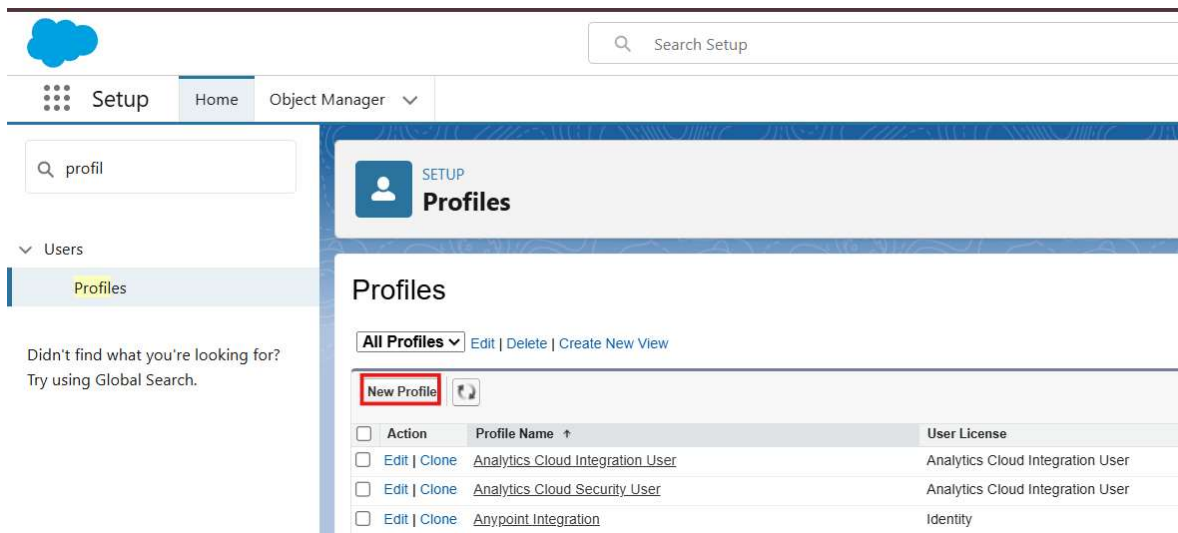
A profile is a collection of settings and permissions controlling what users can access.

Profile Name	Purpose	Main Access
Courier Admin	Full system control	Full CMS object access
Branch Manager	Manage branch operations	Customer/shipment create-read-edit; agent/branch edit

Delivery Agent	Update delivery status	Shipment read-edit; delivery updates create- read-edit
Customer Support	Handle customers and tracking	Customer/shipment read- edit; tracking read

Activity 1: Create a Custom Profile – Courier Admin

- Setup → Profiles → New Profile.
- Existing Profile: System Administrator.



The screenshot shows the Salesforce Setup interface. At the top, there's a search bar labeled 'Search Setup'. Below it, a navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar shows a search bar with 'profil' entered and a list of items including 'Users' and 'Profiles'. The main content area is titled 'Profiles' and includes a 'New Profile' button (highlighted with a red box) and a table of existing profiles.

Action	Profile Name	User License
Edit Clone	Analytics Cloud Integration User	Analytics Cloud Integration User
Edit Clone	Analytics Cloud Security User	Analytics Cloud Integration User
Edit Clone	Anypoint Integration	Identity

- Profile Name: Courier Admin.
- Save.

 **SETUP**
Profiles

Clone Profile

[Help for this Page](#) ?

Enter the name of the new profile.

You must select an existing profile to clone from. | *= Required Information

Existing Profile	System Administrator
User License	Salesforce
Profile Name	Courier Admin

You must select an existing profile to clone from.

Existing Profile	System Administrator
User License	Salesforce

Save

Cancel

Activity 2: Create a Custom Profile – Branch Manager

- Use Standard Platform User as base.
- Name the profile Branch Manager.
- Set CMS app and visible courier tabs.
- Configure required object permissions.
- Save.

Custom Object Permissions

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records 	Modify All Records 	View All Fields
Customer__c	☒	☒	☒	☒	☐	☐	☐
Shipment__c	☒	☒	☒	☒	☐	☐	☐
Delivery_Agent__c	☒	☒	☒	☐	☐	☐	☐
Branch__c	☒	☒	☒	☐	☐	☐	☐
Delivery_Update__c	☒	☒	☒	☒	☐	☐	☐
Invoice__c	☒	☐					

Activity 3: Create a Custom Profile – Delivery Agent

- Use Standard Platform User as base.
- Name the profile Delivery Agent.
- Allow shipment and delivery update access.
- Restrict branch editing and delete permissions.
- Save.

Activity 4: Create a Custom Profile – Customer Support

- Use Standard User as base.
- Name the profile Customer Support.
- Allow customer, shipment and tracking access.
- Save.

Milestone 10- Role

Roles define reporting hierarchy and help determine data visibility.

Activity: Creating Courier Admin Role

- Open Setup → Roles → Set Up Roles.
- Add the role at the required hierarchy level.

Setup Home Object Manager

Profiles
Public Groups
Queues
Roles
User Management Settings
Users

> Data
> Email

PLATFORM TOOLS

> Subscription Management
> Apps
> Feature Settings
> Slack
> Data Cloud
> Heroku
> MuleSoft
> Einstein
> Objects and Fields

SETUP
Roles

Understanding Roles

Set up your Role Hierarchy to control how your organization reports on and accesses data.

Sample Role Hierarchy
View other sample Role Hierarchies: Territory-based Sample

Executive Staff
CEO President CFO VP, Sales

- * View & edit data, roll up forecasts, & generate reports for all users below
- * Can't access data of other Executive Staff

Western Sales Director
Director of W. Sales

Eastern Sales Director
Director of E. Sales

International Sales Director
Director of Int'l Sales

- * View & edit data, roll up forecasts, & generate reports for all users directly below
- * Can't access data of users above or at same level

Western Sales Rep
CA Sales Rep
OR Sales Rep

Eastern Sales Rep
NY Sales Rep
MA Sales Rep

International Sales Rep
Asian Sales Rep
European Sales Rep

- * View & edit data, roll up forecasts, & generate reports only for own data
- * Can't access data of users above or at same level

Set Up Roles

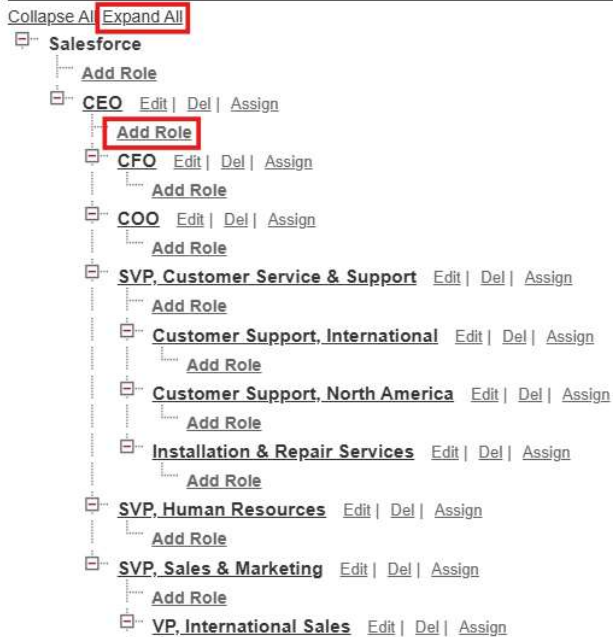
☐ Don't show this page again

- Enter the role label.
- Save.

Creating the Role Hierarchy

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy



Activity: Creating Branch Manager Role

- Open Setup → Roles → Set Up Roles.
- Add the role at the required hierarchy level.
- Enter the role label.
- Save.

Activity: Creating Delivery Agent Role

- Open Setup → Roles → Set Up Roles.
- Add the role at the required hierarchy level.
- Enter the role label.
- Save.

Activity: Creating Customer Support Role

- Open Setup → Roles → Set Up Roles.
- Add the role at the required hierarchy level.
- Enter the role label.
- Save.

Purpose of This Hierarchy: Courier Admin – highest level; Branch Manager – branch operations; Delivery Agent – assigned shipments; Customer Support – customer and tracking support.

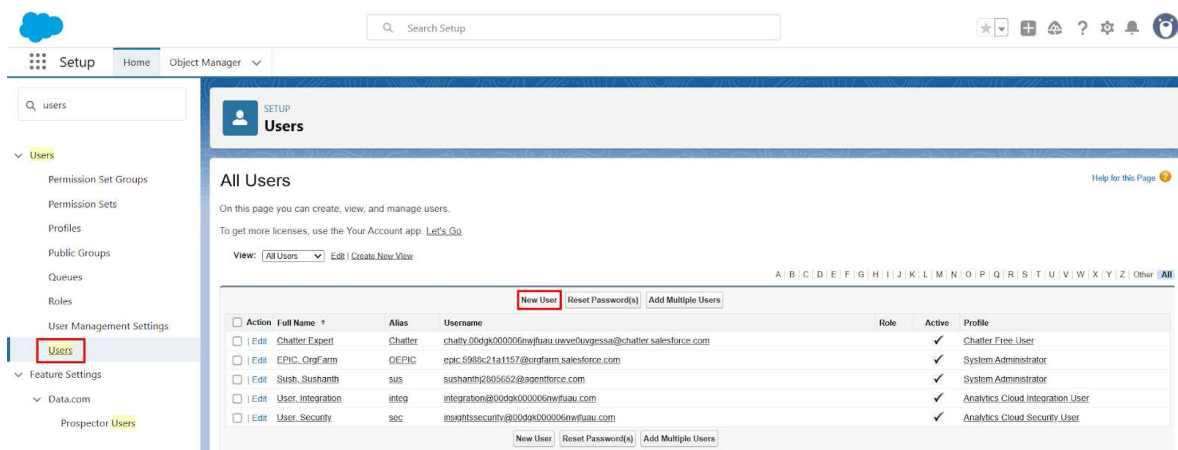
Milestone 11- User

What is a user?

A user is a person who logs in to Salesforce and works with the records and features assigned to that account.

Activity 1: To Create a User

- Setup → Users → New User.
- Enter name and email.
- Create a unique username.



- Assign role and license.
- Assign profile.

- Save.

Activity 2: Creating Other Users

User Type	Role	Profile
Branch Manager	Branch Manager	Branch Manager
Delivery Agent	Delivery Agent	Delivery Agent
Customer Support	Customer Support	Customer Support

Every user should have a unique username, valid email, role and profile. Profiles control permissions and roles support data visibility.

Milestone 15 : Reports & Dashboards

Reports and dashboards help analyze courier data and track shipment activity, agent performance and revenue.

Activity 1: Create Report Types

Report Type	Primary Object	Purpose
Shipment Tracking Report	Shipment__c	Shipment tracking

Customer Summary Report	Customer__c	Customer summary
Agent Performance Report	Delivery_Agent__c	Agent performance

Activity 2: Create Reports

Report 1: Shipment Status Summary Report

- Create Shipment Tracking Report.
- Columns: Tracking Number, Customer, Status, Expected Date, Assigned Agent.
- Filter by required period.
- Group by Status__c and summarize Price.
- Save as Shipment Status Summary.

Report 2: Agent Performance Report

- Create Agent Performance Report.
- Columns: Agent Name, Branch, Number of Shipments, Status.
- Group by Delivery_Agent__c.
- Save and run.

Report 3: Revenue Report

- Create using Shipment Tracking Report.
- Add Tracking Number, Customer, Price and Invoice Status.
- Group by Branch and Month.
- Save as Revenue by Branch Report.

Activity 3: Create Dashboard

- Open Dashboards → New Dashboard.

- Name: Courier Management Dashboard.
- Folder: CMS Reports.
- Create.

Activity 4: Add Dashboard Components

Component	Report	Chart Type
Shipment Status Overview	Shipment Status Summary	Donut Chart
Agent Performance	Agent Performance Report	Bar Chart
Revenue by Branch	Revenue by Branch Report	Column Chart

Activity 5: Save & Activate Dashboard

- Save and finish the dashboard.
- Share with appropriate courier management users.

Final layout:



7. FUNCTIONAL & PERFORMANCE TESTING

Testing verifies record creation, validation rules, delivery status updates, flow automation, invoice information and dashboard visualization.

Test ID	Test Scenario	Expected Result
TC-01	Create Customer record	Valid record is saved
TC-02	Create Shipment record	Tracking number is generated
TC-03	Invalid phone	Validation error appears
TC-04	Zero shipment weight	Save is prevented
TC-05	Past expected date	Validation message appears
TC-06	Delayed status without remarks	Save is prevented
TC-07	Valid delivery update	Update is stored
TC-08	Delivery update status	Shipment status is synchronized
TC-09	Create invoice	Invoice is stored
TC-10	Open dashboard	Charts and reports are displayed

Testing Summary

The test scenarios cover core data entry, validation, automation and reporting functions. Screenshots can be attached as evidence during actual Salesforce implementation.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting
- Structured access control
- Multi-branch support
- Improved tracking visibility

Disadvantages

- Salesforce dependency
- Initial configuration effort
- Some advanced features may require additional setup or licensing
- Users require basic Salesforce training

9. Conclusion

The Courier Management System provides a structured approach to digitizing courier operations through Salesforce. Connecting customer, shipment, delivery agent, branch, delivery update and invoice information supports consistent data management. Validation rules improve data quality, flows reduce repetitive work, profiles and roles provide controlled access, and reports and dashboards support operational review.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

- Customer self-service tracking portal
- Automated delivery-time estimation
- Payment gateway integration

11. APPENDIX

- Source Code / Configuration: Salesforce Configuration
- Dataset: Salesforce Objects
- Demo Environment: Salesforce Developer Org
- Testing Evidence: Salesforce Screenshots

REFERENCES

- Salesforce platform configuration concepts used in the project.
- Salesforce objects, fields, relationships, flows, profiles, roles, reports and dashboards documentation.

Thank You

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2. B. JEEVA BRINDHA
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5. P.MAHALAKSHMI